

Diameter Growth Equations for CONUS with Environment and Climate Variables

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Preliminary CONUS-wide Diameter Growth Equation

Data and Model Form

We built a data set from Forest Inventory and Analysis (FIA) remeasurement data, filtering out observations with missing values for diameter at breast height (**dbh**), basal area in larger trees (**bal**), crown ratio (**cr**), total height (**ht**), site index (**si**), and remeasurement intervals less than 5 years. We also restricted the data to species with ≥ 5000 observations.

The first two principal components of environmental/climatic variables are used. They are:

$$pc_1 = 0.252DD_0 + 0.002TD - 0.035PPT_{sm} + 0.967DD_{18} \quad (1)$$

$$pc_2 = 0.882DD_0 + 0.015TD + 0.42PPT_{sm} - 0.215DD_{18} \quad (2)$$

where:

- DD_0 = growing degree days above $0^\circ C$
- DD_{18} = growing degree days above $18^\circ C$
- TD = Temperature Difference (difference between the mean temperature of the warmest month (MWMT) and the mean temperature of the coldest month (MCMT))
- PPT_{sm} = summer precipitation (mm)

The diameter growth equation fit to these data is:

$$\Delta dbh = e^{(\beta_0 + \beta_1 \log(\frac{(dbh+1)^{\beta_3}}{(cr*ht+1)^{\beta_4}}) + \beta_2 \frac{bal^{\beta_5}}{dbh+2.7} + \beta_5 pc_1 + \beta_6 pc_2)} \quad (3)$$

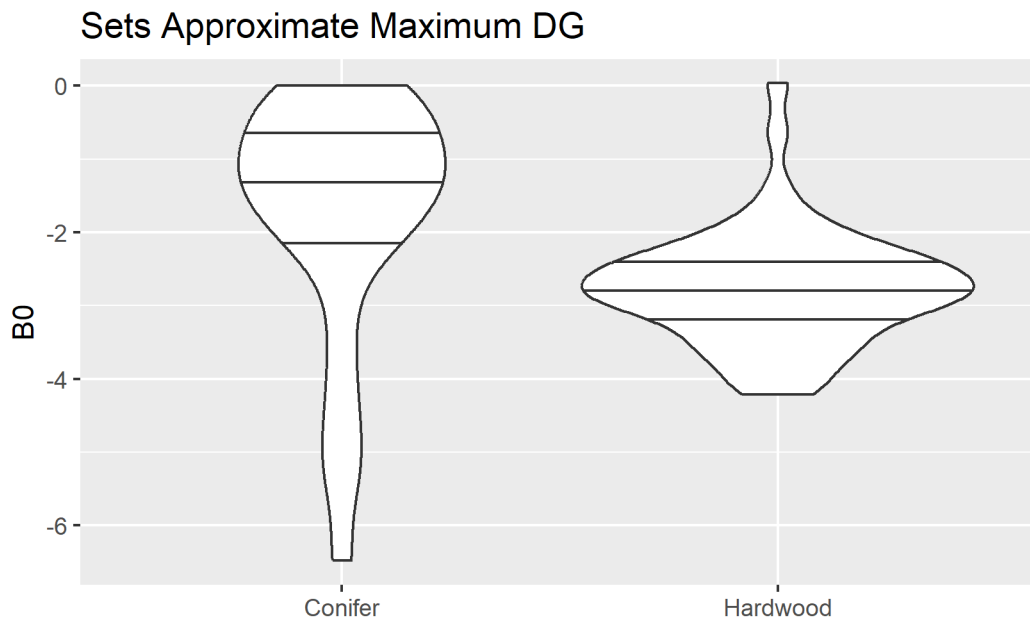
where:

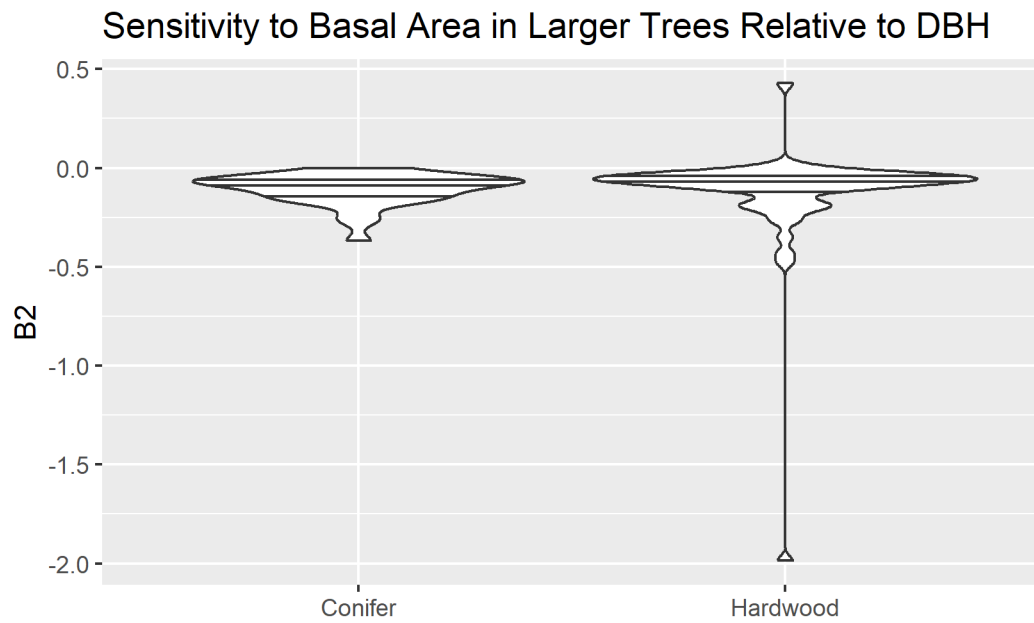
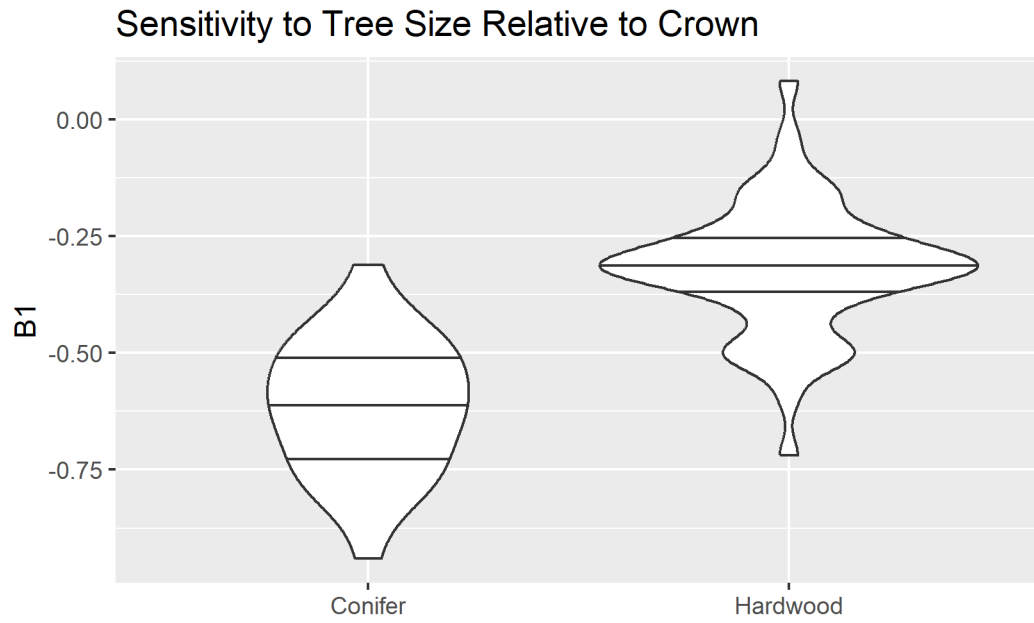
- dbh = diameter at breast height (inches)
- cr = crown ratio (fraction of total height)
- ht = total height (feet)
- bal = basal area in larger trees ($feet^2/acre$)
- pc_1 = environmental/climate principal component 1
- pc_2 = environmental/climate principal component 2

Equation 3 was fit to each qualifying species using an integrated fitting approach and errors minimized were ending dbh . 84 species equations were estimated. Sample sizes ranged from 5099 to 283321 observations.

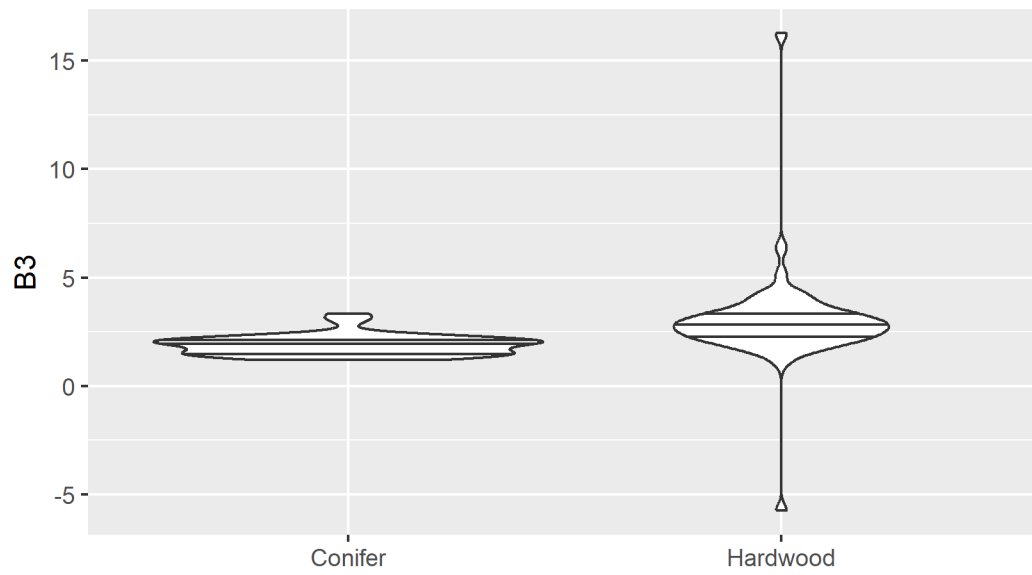
Fitting Results

The graphs below show the range of parameter estimates divided into Conifer and Hardwood groups. The parameter estimates are remarkable stable across species.

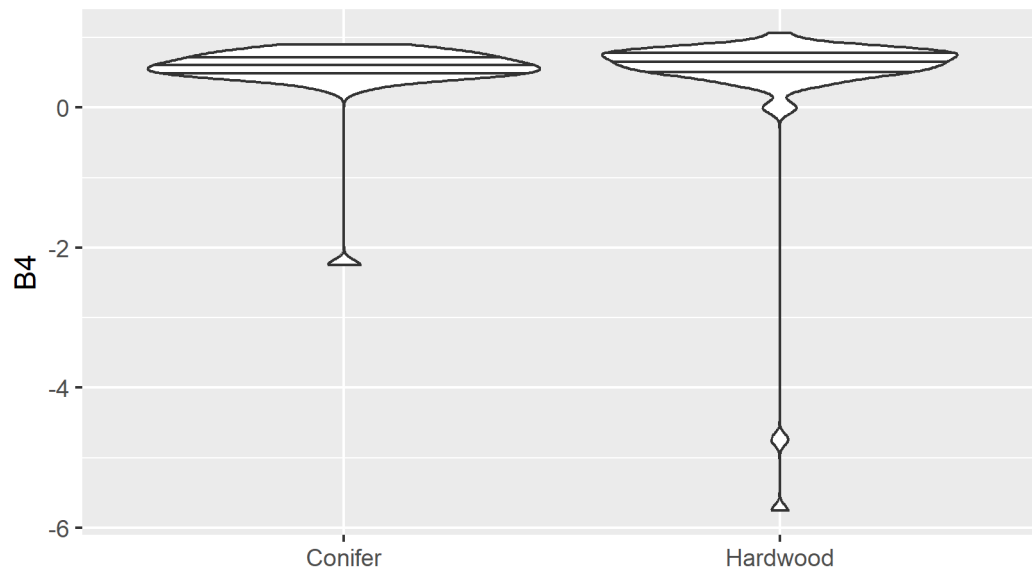


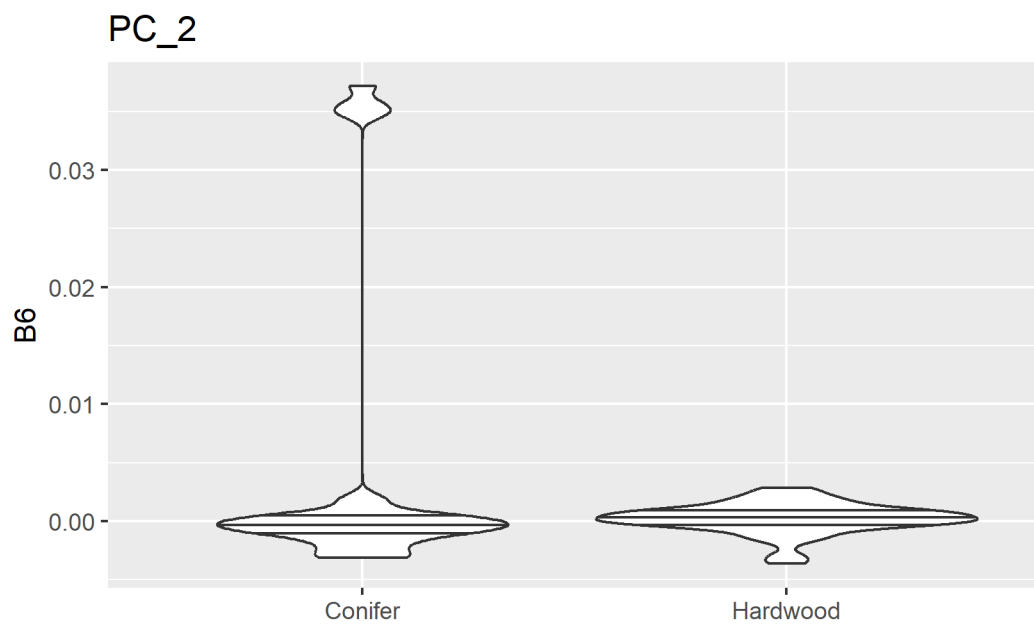
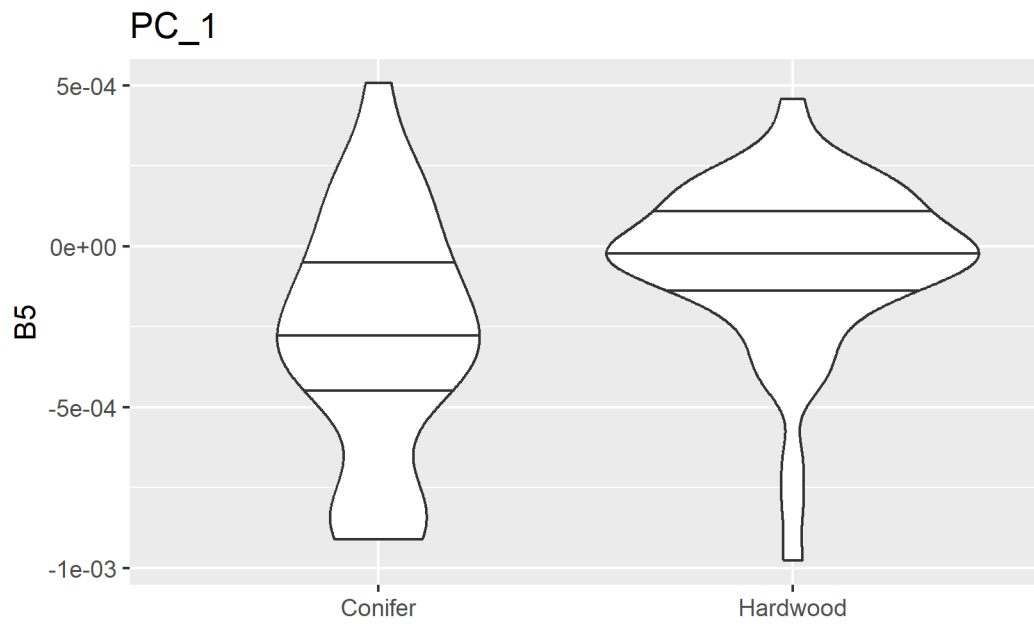


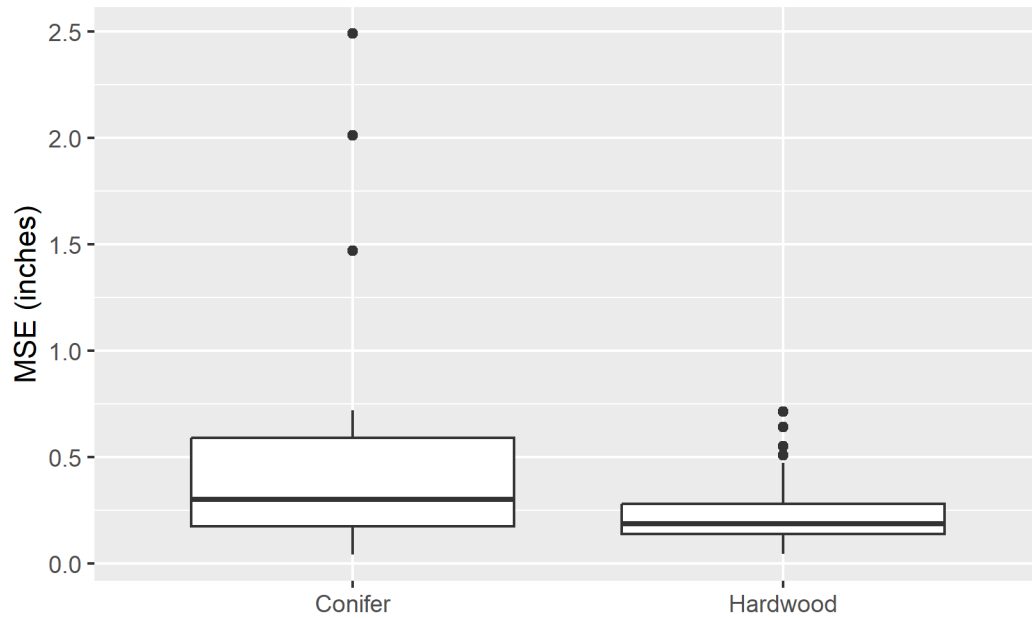
Adjusts Crown Effects



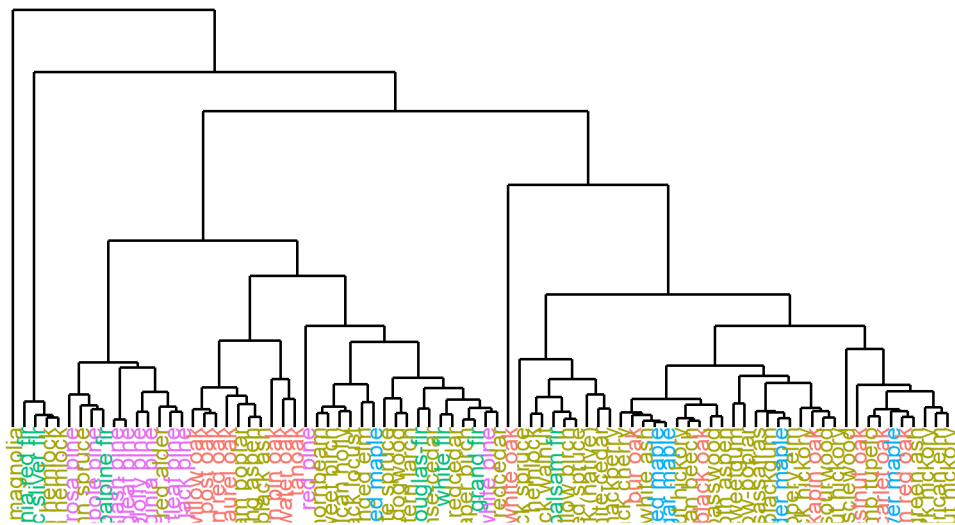
Adjusts Basal Area in Larger Trees Effects



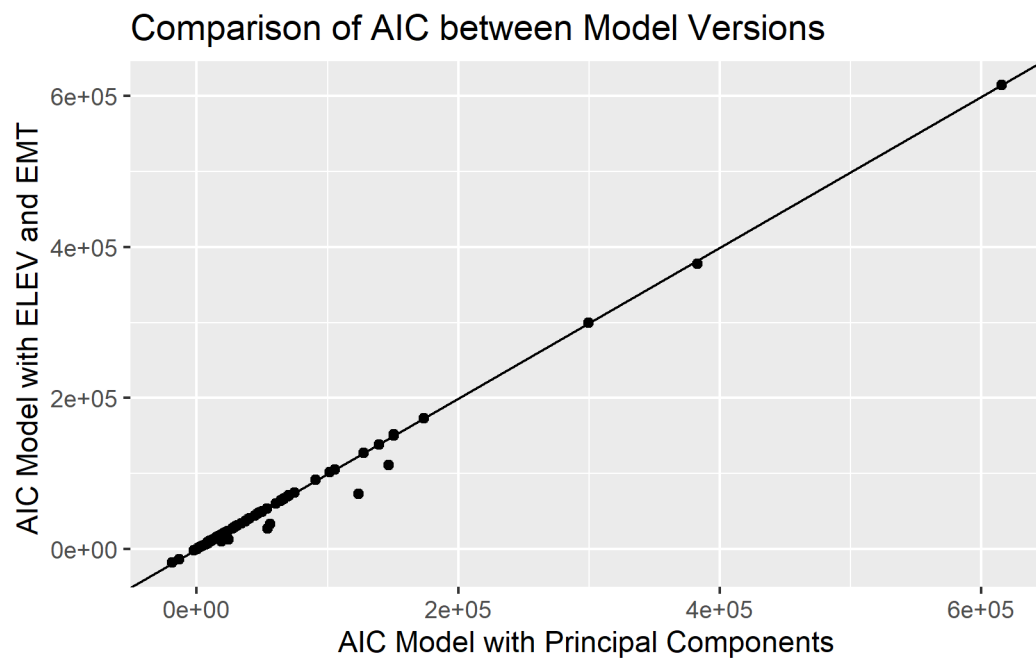




FIA Species Clusters by Diameter Growth Parameter Estimates



Comparison with Previous Model using Elevation and Extreme Minimum Temperature



Residual Analysis

